Simulation of Acoustic Cavitation in Molten NaCl

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Abstract

Ultrasound is used to degas and refine the microstructure of molten metals, and is of growing interest for controlling nucleation (sonocrystallization) in other high-temperature melts, yet the conditions produced inside cavitation bubbles in such melts cannot be measured directly. We present a single-bubble model of acoustic cavitation in molten NaCl at 1100 K, coupling the Keller–Miksis equation to the two-zone heat- and mass-transfer model of Toegel et al. (2000), a covolume-corrected equation of state, and a kinetic scheme for the dimerization, dissociation and ionization of the salt vapor and the argon fill gas. We estimate the effect of the approximations we could test (ionization kinetics, electron heat conduction); for peak temperatures of ∼10 to 25 kK, they produce an uncertainty of ∼15 % (added in quadrature), so the model should be used as an order-of-magnitude estimate for the collapse conditions. The bubble dynamics and heat- and mass-transfer components reproduce the measured maximum radius of a xenon bubble in water to within 3 %. Finally, we use the model to show that for saturation vapor pressures below about 100 Pa the melt vapor changes the peak temperature by less than 10 % at intermediate driving conditions, so a vapor-free version of the model, which requires only the density, viscosity, surface tension and sound speed of the liquid (or approximations if data are unavailable), can be applied to other low-vapor-pressure molten salts and metals.

1 Introduction

When a liquid is subjected to a sufficiently strong sound field, microscopic bubbles within the liquid expand in the low-pressure phase of the acoustic cycle, then collapse during the high pressure phase. This phenomenon is called acoustic cavitation, and it can produce temperatures and pressures on the order of 10^4 K and 10^9 Pa [1, 2]. Acoustic cavitation has applications in medicine, industrial processing, the catalysis of chemical reactions (sonochemistry), crystallization (sonocrystallization) [3], and more.

Ultrasonic processing is not limited to room-temperature liquids. In molten metals it is an established technique for degassing and grain refinement [4, 5], and cavitation bubbles have been imaged directly in an Al–Cu melt by synchrotron radiography [6]. An estimate of the conditions within the bubbles can help inform discussions of possible mechanisms behind the consequences of acoustic cavitation.

In water, the relatively high vapor pressure dampens the collapse. Vapor that enters the bubble during expansion is trapped in the rapid collapse; the endothermic dissociation and added heat capacity of this vapor act as energy sinks [7, 8]. Consistent with this, single-bubble cavitation in liquids of very low vapor pressures such as concentrated sulfuric acid is far hotter and brighter than in water [9], and sonoluminescence has been observed from molten salt eutectics [10]. However, bubble-dynamics modeling of melts has focused on molten metals, mainly on the acoustic field and bubble population [5] and on shape stability [11]. To our knowledge, the interior conditions of an ionic salt melt undergoing acoustic cavitation have not been modeled.

Here, we present a model for acoustic cavitation in NaCl. NaCl was chosen because it is among the best-studied ionic salts; many of its vapor and liquid properties are available in the literature, so relatively few quantities had to be estimated. This model follows a single bubble containing argon and NaCl vapor through multiple acoustic cycles. We have three aims in this work: to explain how the model works, to quantify the uncertainty introduced by modeling approximations and verify (insofar as possible) the model, and to introduce and justify a version of the NaCl simulation that generalizes to any low-vapor pressure

compound. The last item is the most useful in practice. We find that for saturation vapor pressures below about 100 Pa, the vapor changes the peak temperature by less than about 10 % at intermediate driving conditions, so a vapor-free version of the model, requiring only the density, viscosity, surface tension and sound speed of the liquid, can be applied to other low-vapor-pressure molten salts and metals. The model is intended as a semi-quantitative tool: the error sources we can quantify amount to up to about 15 % in peak temperature and pressure, but several model-level assumptions, identified below, are not tested here.

Section 2 outlines the structure of the model and its overarching assumptions. Section 3 gives the equations in the order in which they are evaluated during a timestep. Section 4 justifies the main approximations, estimates their error, and compares the model with measured bubble dynamics in water. Section 5 maps the peak conditions as functions of ambient radius, driving frequency and pressure amplitude, and compares the full model with its vapor-free counterpart. Section 6 summarizes the results and gives practical guidance for applying the model to other systems.

2 Overview of the model

The model describes a single spherical bubble of ambient radius R_0, filled with argon, in molten NaCl at a uniform ambient temperature T_0 and driven by a sinusoidal pressure field of frequency f and amplitude p_A. At each instant the bubble is described by its radius R, wall velocity $\dot{R}$, a single core temperature T, and the number of particles N_i of species i in Table 1.

The radial motion is governed by the Keller–Miksis equation [12], which gives the wall acceleration $\ddot{R}$ from the gas pressure and its rate of change. The gas pressure follows from a covolume-corrected equation of state, so the central task at each timestep is to determine the gas temperature and composition. These are set by three coupled processes: heat conduction and vapor exchange with the liquid; chemical reactions in the gas (dimerization, dissociation and ionization of the salt vapor, and ionization of argon); and the first law of thermodynamics, which combines compression work, heat loss, the enthalpy carried by vapor, and the energy absorbed or released by reactions into a single equation for $\dot{T}$.

Heat and mass transfer are treated with the two-zone model of Toegel et al. [7]. The bubble is divided into a homogeneous core and a thin boundary layer, across which temperature and composition are interpolated between their core values and their values at the wall. The thickness of this layer is set by the diffusion length over the timescale of the bubble motion, so it is thin during rapid collapse and thick during slow expansion (Section 3.4).

The model rests on the following overarching assumptions:

- The bubble remains spherical. Strongly driven bubbles become shape-unstable (Section 4.6), but below that threshold spherical models give useful estimates of peak conditions.
- The pressure inside the bubble is spatially uniform, and the liquid at the wall remains at T_0, so the density, viscosity, surface tension and sound speed of the liquid can be held constant. Both assumptions are supported by simulations of the full fluid-dynamical equations [13].
- The non-condensable gas is argon, which neither dissolves in nor reacts with the melt.
- The vapor concentration at the wall equals the saturation concentration of the liquid at T_0.
- The bubble is isolated: interactions with other bubbles and with the container, and spatial variation of the acoustic field, are not considered.

Unless otherwise stated, all simulation runs described in this document use the default parameters frequency $f = 20\,\mathrm{kHz}$, ambient radius $R_0 = 5\,\mu\mathrm{m}$, pressure amplitude $p_\mathrm{A} = 1.2\,\mathrm{atm}$, and ambient temperature $T_0 = 1100\,\mathrm{K}$.

3 Equations

3.1 Equation of state

The volume-corrected ideal gas law is used for this simulation:

$$p_{\mathrm{g}}(V - B) = Nk_{\mathrm{B}}T \tag{1}$$

where p_{g}, V, N and T are the pressure, volume, total number of particles and core temperature of the gas in the bubble, N_i is the number of particles of species i, N_{sp} is the number of species, and $B = \sum_{i=1}^{N_{\mathrm{sp}}} N_i b_i$ is the total volume occupied by the gas molecules themselves.

For species without measured b_i values (e.g., Na, Cl), we use the approximate formula $b_i = (16/3)\pi r_i^3$, with r_i being the van der Waals radius of species i [14]. For larger molecules we simply add the covolumes of the constituent atoms to produce a rough estimate of the overall covolume.

3.2 Caloric properties

The total heat capacity is approximated using the equation

$$C_v = \frac{3}{2}Nk_{\mathrm{B}} + \sum_{i=1}^{N_{\mathrm{sp}}} N_i c_{\mathrm{nt},i} \tag{2}$$

where $(3/2)Nk_{\mathrm{B}}$ represents the translational modes (which every species has) and $c_{\mathrm{nt},i}$ represents the heat capacity per molecule from any non-translational modes of energy storage of species i. The non-translational modes include rotation, vibration, and electronic excitation:

$$c_{\mathrm{nt},i} = c_{\mathrm{rot},i} + c_{\mathrm{vib},i} + c_{\mathrm{elec},i}$$

3.2.1 Rotation

Of the species in the model, only NaCl and Na_2Cl_2 have rotational and vibrational modes. By equipartition, $c_{\mathrm{rot,NaCl}} = k_{\mathrm{B}}$ and $c_{\mathrm{rot,Na_2Cl_2}} = (3/2)k_{\mathrm{B}}$ [15].

3.2.2 Vibration

Similarly, energy is only stored in vibrational modes for NaCl (linear, two atoms, so one vibrational mode) and Na_2Cl_2 (nonlinear, four atoms, so six vibrational modes). From statistical mechanics [15], we have

$$c_{\mathrm{vib}} = k_{\mathrm{B}} \sum_{i=1}^{N_{\mathrm{vib}}} \left(\frac{\theta_i}{T}\right)^2 \frac{e^{\theta_i/T}}{\left(e^{\theta_i/T} - 1\right)^2}$$

where $\theta_i = hc\tilde{\nu}_i/k_{\mathrm{B}}$ (with $\tilde{\nu}_i$ being the vibrational frequency of mode i) is the characteristic vibrational temperature of mode i. The characteristic vibrational frequencies have been calculated for both NaCl and Na_2Cl_2 [16, 17].

3.2.3 Electronic excitation

Electronic excitation is irrelevant for Na^+ and Cl^- due to their closed-shell configurations. For NaCl and Na_2Cl_2, excitation is neglected due to a lack of electronic data for the two compounds. This is unlikely to significantly affect the accuracy of the simulation, as discussed in Section 4.1. For the other species, we again have from statistical mechanics [15]

$$c_{\mathrm{elec}} = \frac{k_{\mathrm{B}}}{T^2}\left[\frac{\sum_i \theta_i^2 g_i e^{-\theta_i/T}}{\sum_i g_i e^{-\theta_i/T}} - \left(\frac{\sum_i \theta_i g_i e^{-\theta_i/T}}{\sum_i g_i e^{-\theta_i/T}}\right)^2\right]$$

where θ_i is defined analogously to Section 3.2.2 and g_i represents the degeneracy of state i. Electronic level data are sourced from the NIST Atomic Spectra Database pages of the relevant ions and atoms [18].

3.3 Transport properties

The thermal conductivity of the gas mixture is approximated as the product of a dilute-gas conductivity $\kappa_0(T)$ and a dense-gas enhancement factor F_E:

$$\kappa(T, n_\mathrm{h}) = \kappa_0(T)\, F_\mathrm{E}(T, n_\mathrm{h}), \qquad \kappa_0(T) = \frac{1}{N_\mathrm{h}} \sum_{i=1}^{N_\mathrm{sp}} \left(N_i \kappa_i(300)\right) \left(\frac{T}{300}\right)^{0.7} \tag{3}$$

with N_h being the number of heavy particles in the bubble, $\kappa_i(300)$ being the thermal conductivity of species i at 300 K, and the 0.7 exponent being an approximate empirical parameter [19]. κ_0 is then the (empirical) thermal conductivity of the mixture in the dilute-gas limit, and F_E is the Enskog dense-gas enhancement, evaluated for pure argon since the mixture is $\gtrsim 90\,\%$ argon for the parameter sets considered here. Following Bastea [20],

$$F_\mathrm{E} = \frac{1}{g_c} + \frac{6}{5} y + 0.7574\, y^2 g_c, \qquad y = \tfrac{2}{3}\pi n_\mathrm{h} d^3, \qquad g_c = \frac{1 - y/8}{(1 - y/4)^3}, \tag{4}$$

where $n_\mathrm{h} = N_\mathrm{h}/V$, g_c is the Carnahan–Starling contact value [21], and $d(T)$ is the Barker–Henderson effective diameter [22] computed from the exponential-6 argon potential of Ref. [20]. $F_\mathrm{E} \to 1$ in the dilute limit. Unlike Ref. [20], F_E multiplies the empirical $\kappa_0(T)$ rather than the hard-sphere dilute conductivity. Ref. [20] validates the method up to 3000 K, so its use at collapse temperatures is an extrapolation.

Two further assumptions are made. First, the species' conductivities are combined as a number-weighted average. This is not exact, but the conductivities are of the same order of magnitude and the gas is predominantly argon, so the result is close to that of pure argon regardless. Second, electron heat conduction is neglected (Section 4.4). The thermal conductivity can then be used to calculate the thermal diffusivity χ of the mixture

$$\chi = \frac{\kappa}{\frac{N}{V}(c_v + k_\mathrm{B})} \tag{5}$$

where $c_v = C_v/N$ is the average heat capacity per particle. This plays a role in Section 3.4.

The diffusivity of species i, D_i, can be shown to be proportional to $\sqrt{T}/n$ in the hard-sphere approximation [23]:

$$D_i = D_{0,i} \frac{n_\mathrm{ref}}{n} \sqrt{\frac{T}{T_\mathrm{ref}}} \tag{6}$$

where $D_{0,i}$ is the diffusivity of species i in argon at $T_\mathrm{ref} = 1100$ K and 1 atm, with n_ref the corresponding number density. The binary diffusivities in argon D_0 and thermal conductivities κ of NaCl, Na_2Cl_2, Na and Cl were estimated from Chapman–Enskog theory [24, 25], using Lennard-Jones parameters from [26] and collision integrals from [27]. Heat capacities for the thermal conductivity estimates were taken from [28]. Equation 6 assumes that the molecules are diffusing through pure argon, which is an acceptable simplification given that the mixture is predominantly argon throughout the simulation.

3.4 Boundary layer thickness

The liquid is treated as an isothermal reservoir at T_0 with saturated vapor at the wall, so the resistance to heat and mass transfer lies in a thin layer on the gas side of the interface. As discussed in Section 2, we use the two-zone model of Ref. [7], which employs several characteristic boundary layer thicknesses. The first is for heat:

$$l_\mathrm{t} = \min\left(\sqrt{\frac{\chi R}{|\dot{R}|}}, \frac{R}{\pi}\right) \tag{7}$$

where l_t is the thermal diffusion boundary layer thickness and χ is the thermal diffusivity of the gas mixture (Section 3.3). At low $\dot{R}$, this is capped by R/π, as shown by Toegel et al. [7]. A similar boundary layer thickness is calculated for particle diffusion:

$$l_i = \min\left(\sqrt{\frac{D_i R}{\left|\dot{R}\right|}}, \frac{R}{\pi}\right) \tag{8}$$

where l_i is the diffusion boundary layer thickness for species i and D_i is the diffusivity of species i. Both follow from the diffusion length $\sqrt{Dt}$ (or $\sqrt{\chi t}$) over the timescale of the bubble wall's motion, $t \sim R/|\dot{R}|$, so the layer is thin during rapid collapse and thick during slow expansion.

3.5 Heat conduction and vapor exchange

The two-zone model of Ref. [7] gives the rate of heat conduction to the liquid as

$$\dot{Q} = \frac{4\pi\kappa R^2}{l_\mathrm{t}}(T_0 - T) \tag{9}$$

which is then considered in Section 3.9 to find the rate of change of the bubble temperature. The rate at which species i diffuses into the bubble is

$$\dot{S}_i = \frac{4\pi D_i R^2}{l_i}(n_{\mathrm{sat},i} - n_i) \tag{10}$$

where $n_{\mathrm{sat},i}$ and n_i represent the saturation vapor concentration and current vapor concentration of species i, and $\dot{S}_i$ represents the diffusion rate in particles s^{-1}. The thermochemistry of this exchange (and thus its impact on $\dot{T}$) is further discussed in Section 3.9.

3.6 Ambipolar diffusion

When electrons diffuse out of the bubble, it acquires a net charge, producing an electric field that pushes an equivalent number of oppositely charged particles out (here, positively charged ions) and holds back the electrons. This is called ambipolar diffusion, in which charged particles diffuse in oppositely charged pairs. Generally, one of the particles is an electron and the other is a positively charged ion. In this situation, the negatively charged particle may also be a Cl^- ion, but N_{e^-} is more than two orders of magnitude higher than N_{Cl^-} during the collapse, so here we assume that the negatively charged particle is almost always an electron. The pairs then diffuse according to Equation 10, with effective diffusion coefficient [29]

$$D_\mathrm{a} \approx 2D_\mathrm{ion} \tag{11}$$

where D_ion is assumed to be the same as the diffusivity of the parent neutral atom, which is valid as an order-of-magnitude estimate under the conditions of this simulation [30, 31, 32]. As discussed in Section 4.3, the timescale of ambipolar diffusion and its nature as a post-collapse process mean that any results of interest (namely, peak temperature and pressure during the collapse) are not meaningfully changed by variation in the rate of ambipolar diffusion, so an order-of-magnitude estimate is acceptable. Equation 10 can then be applied to each of Ar^+, Na^+, and Cl^+, with e^- and Cl^- assumed to be paired to positive ions proportionally to their concentrations. The validity of this assumption is also discussed in Section 4.3.

Argon ions reaching the wall by ambipolar diffusion are assumed to be neutralized there by charge transfer from the melt and to return to the bubble as neutral atoms, with the recombination energy deposited in the liquid. Argon is assumed not to dissolve in the melt, as noble-gas solubilities in molten salts are extremely low.

3.7 Equilibrium constants

We have from statistical mechanics [15] that the equilibrium constant for a chemical reaction is given by the product of the constituent partition functions per unit volume:

$$K_\mathrm{eq} = \prod_i \left(Q_i^{\nu_i}\right) \exp\left(-\frac{\Delta U}{k_\mathrm{B}T}\right) \tag{12}$$

#	Reaction	Notes
1	$2\,\mathrm{NaCl} \leftrightarrow \mathrm{Na_2Cl_2}$	
2	$\mathrm{NaCl + M} \leftrightarrow \mathrm{Na + Cl + M}$	M is any heavy particle.
3	$\mathrm{NaCl + M} \leftrightarrow \mathrm{Na^+ + Cl^- + M}$	M is any heavy particle.
4	$\mathrm{Na} \leftrightarrow \mathrm{Na^+ + e^-}$	
5	$\mathrm{Cl + e^-} \leftrightarrow \mathrm{Cl^-}$	
6	$\mathrm{Cl} \leftrightarrow \mathrm{Cl^+ + e^-}$	
7	$\mathrm{Na^+ + Cl^-} \leftrightarrow \mathrm{Na + Cl}$	Faster pathway for $\mathrm{Na^+/Cl^-}$ neutralization than the reverse of Reaction 3.
8	$\mathrm{Cl^- + Cl^+} \leftrightarrow 2\,\mathrm{Cl}$	Lets $\mathrm{Cl^-}$ and $\mathrm{Cl^+}$ neutralize each other when no $\mathrm{Na^+}$ or electrons are available.
9	$\mathrm{Ar} \leftrightarrow \mathrm{Ar^+ + e^-}$	
10	$\mathrm{Ar^+ + Cl^-} \leftrightarrow \mathrm{Ar + Cl}$	Necessary for the same reason as Reaction 8.

Table 1: Reactions considered in the simulation

where Q_i and ν_i represent the partition functions and stoichiometric coefficients, respectively, of the ith species involved in the reaction, and ΔU represents the change in internal energy of the reaction. We also have that the partition function of a particular species is given by [15]

$$Q = q_{\mathrm{trans}} q_{\mathrm{rot}} q_{\mathrm{vib}} q_{\mathrm{elec}}$$

with

$$q_{\mathrm{trans}} = \left(\frac{\sqrt{2\pi m k_{\mathrm{B}} T}}{h} \right)^3$$

$$q_{\mathrm{rot,\ lin}} = \frac{T}{\sigma \theta_{\mathrm{rot}}}$$

$$q_{\mathrm{rot,\ nonlin}} = \frac{\sqrt{\pi T^3}}{\sigma \sqrt{\theta_A \theta_B \theta_C}}$$

$$q_{\mathrm{vib}} = \prod_{i=1}^{N_{\mathrm{vib}}} \frac{1}{1 - \exp\left(-hc\tilde{\nu}_i / k_{\mathrm{B}} T\right)}$$

$$q_{\mathrm{elec}} = \sum_{i=1}^{N_{\mathrm{levels}}} g_i \exp\left(-\epsilon_i / k_{\mathrm{B}} T\right)$$

where m is the mass of the particle, ϵ_i represents the energy of the ith electronic level, σ is the symmetry number for the molecule, and θ_{rot} and $\theta_{\mathrm{A,B,C}}$ are the characteristic rotational temperatures of the molecule [16, 33, 34]. When applied to ionization, Equation 12 becomes a form of the Saha ionization equation:

$$K_{\mathrm{eq}} = \frac{n_{\mathrm{e^-}} n_{\mathrm{positive}}}{n_{\mathrm{neutral}}} = \frac{Q_{\mathrm{e^-}} Q_{\mathrm{p}}}{Q_{\mathrm{n}}} \exp\left(-\frac{E_{\mathrm{ion}}}{k_{\mathrm{B}} T}\right) \tag{13}$$

where the partition functions of the species (Q_{p} for the positive ion and Q_{n} for the neutral) are calculated using the formulas above and the ionization energies are available on the NIST Atomic Spectra Database [18]. This equation simply gives the equilibrium ion composition at a particular temperature. Second ionization is not considered in this simulation, and while it is unlikely to matter significantly at temperatures under $\sim 25\,\mathrm{kK}$ (judging by the ionization fraction of singly charged ions at these temperatures), it may become more significant above.

3.8 Rate constants

Forward rate constants are estimated from collision theory [35],

$$k_\mathrm{f} = \pi d^2 \sqrt{\frac{8 k_\mathrm{B} T}{\pi \mu_{AB}}} \exp\left(-\frac{E_a}{k_\mathrm{B} T}\right) \tag{14}$$

where μ_{AB} represents reduced mass. A common collision diameter $d = 5$ Å is used, and the activation energy E_a is taken to be equal to the reaction endothermicity. For heavy-particle impact ionization (Reactions 4, 6 and 9), k_f is multiplied by an efficiency factor of 0.01, reflecting the suppression of electronic transitions in slow heavy-particle collisions [36]; the resulting uncertainty is bracketed in Section 4.2. Equation 14 does not take into account the orientation of the molecules, but as most of the species considered are either atoms or two-atom molecules, the impact of this approximation is limited. The rate of the reverse reaction is then calculated as $k_\mathrm{r} = k_\mathrm{f}/K_\mathrm{eq}$ to ensure convergence to equilibrium.

Under the conditions present here ($T \sim 10^4$ K), both heavy-particle impact ionization (HPII) and electron impact ionization (EII) contribute to the ionization rate [37]. EII is neglected here, with an error analysis in Section 4.2. The chemical reaction for HPII is

$$\mathrm{Na} + \mathrm{M} \leftrightarrows \mathrm{Na^+} + \mathrm{e^-} + \mathrm{M}$$

with rate constant assumed to be given by Equation 14, and an analysis of the uncertainty created by this assumption in Section 4.2.

3.9 Energy equation

For clarity, in this section, i indices will refer to species i, and j indices to reaction j. Then, the total energy in the bubble is given by

$$U = \sum_i N_i \left(E_i^0 + \frac{3}{2} k_\mathrm{B} T + \int_0^T c_{\mathrm{nt},i}(T')\,\mathrm{d}T'\right) = \sum_i N_i u_i(T)$$

where E_i^0 is the 0 K energy of species i relative to separated neutral atoms and $u_i(T) \equiv E_i^0 + (3/2)k_\mathrm{B}T + \int_0^T c_{\mathrm{nt},i}(T')\,\mathrm{d}T'$ is the energy of a single molecule of species i at temperature T. Taking the time derivative, we find

$$\dot{U} = \sum_i \left(N_i \left(\frac{3}{2} k_\mathrm{B} \dot{T} + c_{\mathrm{nt},i} \dot{T}\right) + \dot{N}_i u_i(T)\right)$$

and thus

$$\dot{U} = C_v \dot{T} + \sum_i \dot{N}_i u_i(T) \tag{15}$$

From the first law of thermodynamics, we also have

$$\dot{U} = \dot{Q} - p_\mathrm{g} \dot{V} + \sum_i \dot{S}_i h_i \tag{16}$$

where h_i is the enthalpy per particle of species i entering or leaving the bubble, and the third term represents the energy carried by molecules entering or leaving the bubble. Enthalpy is used rather than internal energy in order to account for the work done in pushing the molecules into or out of the bubble. Setting Equation 16 equal to Equation 15, we have

$$C_v \dot{T} + \sum_i \dot{N}_i u_i(T) = \dot{Q} - p_\mathrm{g} \dot{V} + \sum_i \dot{S}_i h_i$$

Now we define $\dot{W}_i$ as the rate at which molecules of species i are produced via the chemical reactions in Table 1. Then, $\dot{N}_i = \dot{S}_i + \dot{W}_i$. Thus, we have

$$\dot{T} = \frac{\dot{Q} - p_\mathrm{g} \dot{V} + \sum_i \dot{S}_i \left(h_i - u_i(T)\right) - \sum_i \dot{W}_i u_i(T)}{C_v}$$

Finally, $\sum_i \dot{W}_i u_i(T)$ is simply the total rate of change of chemical energy. This can be rewritten as $V \sum_j r_j(T) \Delta U_j(T)$ where $r_j(T)$ is the rate of reaction j per m^3 and $\Delta U_j(T)$ is the total change in internal energy of reaction j per single reaction. Therefore, the final formula for $\dot{T}$ is

$$\dot{T} = \frac{\dot{Q} - p_\mathrm{g}\dot{V} + \sum_i \dot{S}_i \left(h_i - u_i(T)\right) - V \sum_j r_j(T) \Delta U_j(T)}{C_v} \tag{17}$$

3.10 Keller–Miksis equation

The Keller–Miksis equation is given by [12]

$$\left(1 - \frac{\dot{R}}{c}\right) R\ddot{R} + \frac{3}{2}\dot{R}^2 \left(1 - \frac{\dot{R}}{3c}\right) = \left(1 + \frac{\dot{R}}{c}\right) \frac{p_\mathrm{l}(t) - p_\mathrm{ac}(t) - P_0}{\rho_\mathrm{l}} + \frac{R}{\rho_\mathrm{l} c} \frac{\mathrm{d}p_\mathrm{l}}{\mathrm{d}t} \tag{18}$$

where R is the bubble radius, $\dot{R}$ and $\ddot{R}$ are the velocity and acceleration of the bubble wall, c and ρ_l are the speed of sound in and density of the liquid, $P_0 = 1\,\mathrm{atm}$ is the static pressure in the liquid, and $p_\mathrm{ac}(t) = -p_\mathrm{A} \sin(2\pi f t)$ is the acoustic driving pressure, with amplitude p_A and frequency f. In the simulation, we use this equation to solve for $\ddot{R}$ and determine how the bubble evolves (density, speed of sound, surface tension, and viscosity of NaCl from Refs. [38, 39]). To do this, we must find p_l, the pressure on the liquid side of the bubble interface. Because of surface tension and viscosity, this differs from the gas pressure p_g. From the Young–Laplace equation [40], we have that the pressure difference across the interface due to surface tension is

$$\Delta p_\sigma = -\frac{2\sigma}{R} \tag{19}$$

for a spherical interface, where σ is the surface tension of the liquid, with the sign being negative because the liquid is on the convex side of the interface. In a spherical geometry, the pressure difference across the interface due to viscosity is given by [41]

$$\Delta p_\mu = -\frac{4\mu\dot{R}}{R} \tag{20}$$

where μ is the dynamic viscosity of the liquid, with the sign being negative because viscosity acts against the direction of motion. Thus, using Newton's second law on the (massless) boundary between the liquid and the gas, we have that $p_\mathrm{l} = p_\mathrm{g} + \Delta p_\sigma + \Delta p_\mu$ and thus

$$p_\mathrm{l} = p_\mathrm{g} - \frac{2\sigma}{R} - \frac{4\mu\dot{R}}{R} \tag{21}$$

To calculate $\mathrm{d}p_\mathrm{l}/\mathrm{d}t$ for use in Equation 18, we simply take the derivative of Equation 21, substituting in Equation 1:

$$\frac{\mathrm{d}p_\mathrm{l}}{\mathrm{d}t} = \frac{k_\mathrm{B}}{V - B} \left(\dot{N}T + N\dot{T} - \frac{NT\left(\dot{V} - \dot{B}\right)}{V - B} \right) + \frac{2\sigma\dot{R}}{R^2} - \frac{4\mu\ddot{R}}{R} + \frac{4\mu\dot{R}^2}{R^2} \tag{22}$$

which can then be substituted into Equation 18 to solve for $\ddot{R}$.

3.11 Numerics

The integrator used for this simulation is SciPy's Radau, which is an implicit adaptive timestepping integrator. A convergence study was run, varying relative tolerance (rtol) and measuring peak temperature, with the results displayed in Figure 1. Accordingly, a relative tolerance of 1×10^{-8} was selected, as at this tolerance the peak temperature converges to well within 0.01 %, with further tightening producing negligible change, while the run time and computational cost remain reasonable.

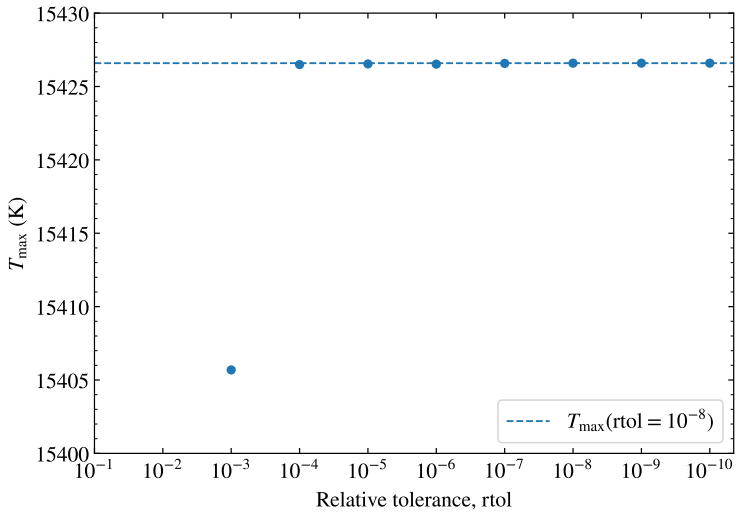
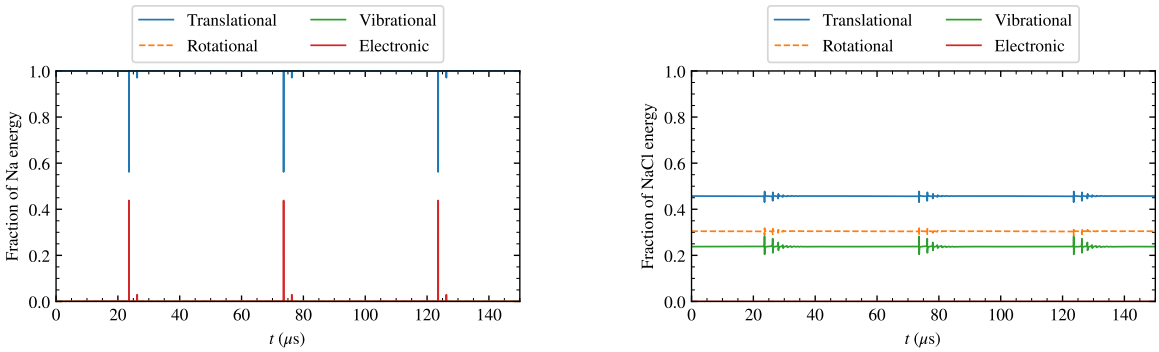


Figure 1: Convergence of peak temperature with decreasing relative tolerance; default parameters were used (Sec. 2).



(a) Na

(b) NaCl

Figure 2: Proportion of thermal energy stored in translational, rotational, vibrational, and electronic modes. Physically, NaCl's electronic excitation energy is not zero during the collapse, but the simulation neglects it.

4 Justification of assumptions and quantification of error

4.1 Neglected heat capacity due to electronic excitation

Due to a lack of available data or a simple method for approximation, this simulation neglects the energy stored in NaCl and Na_2Cl_2 by electronic excitation. As seen in Figure 2, the energy stored by electronic excitation is only non-negligible during the collapse. Because NaCl and Na_2Cl_2 are not present in any appreciable quantity at this time (as they dissociate into Na and Cl), neglecting their electronic excitation does not meaningfully affect the results.

4.2 Ionization rate uncertainty

Electron impact ionization (EII) is neglected in this simulation. At high temperatures, EII's contribution to the ionization rate often outweighs HPII's contribution [42]. Additionally, collision theory rates are quite unreliable for HPII, since electronic transitions in slow heavy-particle collisions are strongly suppressed and classical estimates of their rates can err by orders of magnitude [36].

Figure 3 brackets (as a function of pressure amplitude) the error in peak temperature and pressure created by the aforementioned ionization rate inaccuracies. For $p_A = 1.2\,\text{atm}$, the bounds on the peak temperature T_{max} are approximately $\delta T^+_{max} = 1\,\%$ and $\delta T^-_{max} = -5\,\%$. In general, these bounds increase for parameter sets yielding higher peak temperatures (high p_A, low f). For such parameter sets, the bubble is expected to approach or exceed shape-stability thresholds [1], which are not computed here. In this regime, the spherical model itself, rather than the ionization rate, is likely the main limitation.

4.3 Timescale of ion recombination and ambipolar diffusion

Electrons are far more mobile than Cl^-, so in reality they probably leave the bubble faster than their share of the negative charge, which the model does not capture. However, ambipolar diffusion removes the charge within about 0.5 µs (Figure 4), after the temperature peak and long before the next collapse ($> 10\,\mu s$ later), so this does not affect the peak conditions.

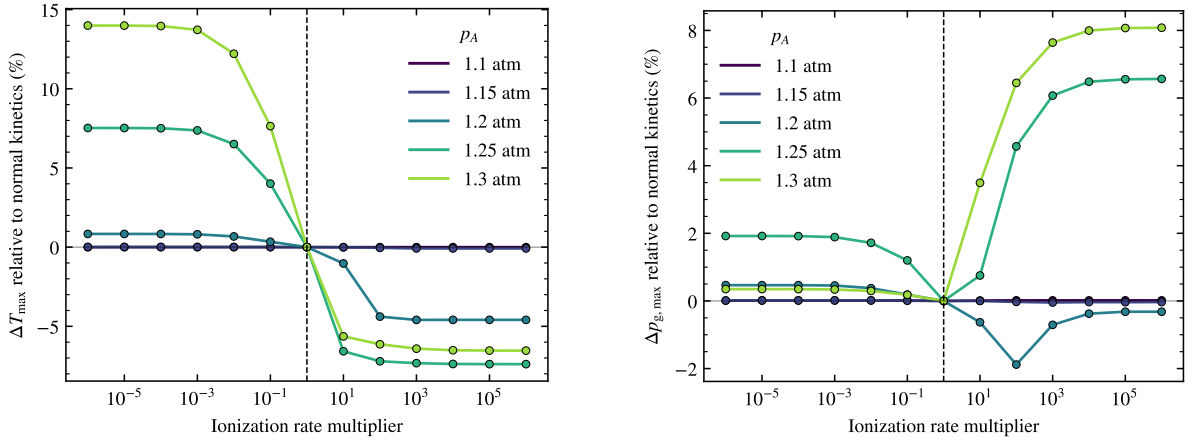
In the model, positive ions recombine with electrons only through the reverse of heavy-particle impact ionization (three-body recombination with a heavy third body) and through the mutual-neutralization Reactions 7, 8 and 10. Recombination with an electron as the third body, which dominates at the low temperatures after the collapse, is omitted, so recombination in the model is too slow. Instead, the remaining charge leaves the bubble by ambipolar diffusion within $\sim 0.5\,\mu s$ (Figure 4). Either route removes the charge long before the next collapse, so this does not affect the peak conditions.

4.4 Thermal conductivity error

The model neglects heat conduction by electrons. Outside the collapse the gas is essentially un-ionized, so this matters only near the temperature peak.

To bound the error introduced by this approximation, we estimate the electron thermal conductivity from elementary kinetic theory [43], scaling the heavy-particle conductivity by the electron fraction and the ratio of thermal speeds, and assuming comparable electron and heavy-particle mean free paths:

$$\kappa_{e^-} \approx \frac{N_{e^-}}{N_h}\sqrt{\frac{m_h}{m_{e^-}}}\kappa_{h,\,\text{dilute}} \tag{23}$$



(a) Peak temperature. At an ionization rate multiplier of 1, the 1.2 and 1.3 atm trials have peak temperatures of about 15.4 and 25.8 kK, respectively.

(b) Peak pressure. At an ionization rate multiplier of 1, the 1.2 and 1.3 atm trials have peak pressures of about 0.82 and 4.5 GPa, respectively.

Figure 3: Variation of peak temperature and pressure with ionization rate multiplier and pressure amplitude. The far-right side of each plot (large ionization rate multiplier) is the limit of fast ionization (i.e., the system remains in Saha equilibrium). The far-left side of each plot is the limit of no ionization.

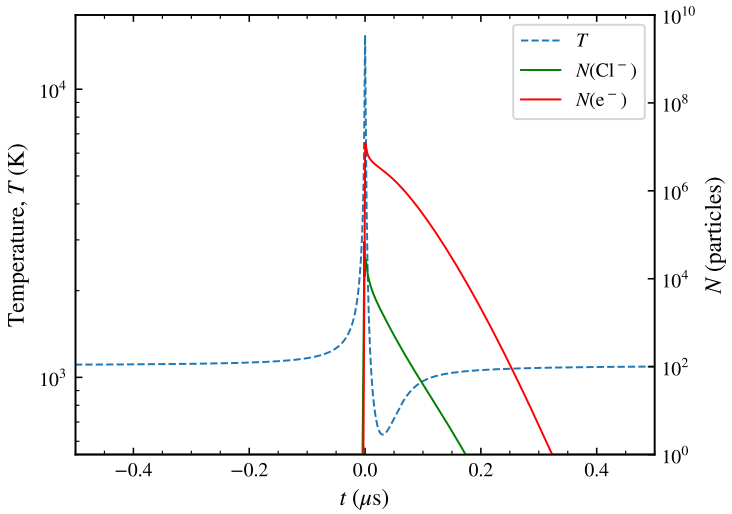


Figure 4: T, N_{Cl^-}, and N_{e^-} over time near the collapse. The timescale of ambipolar diffusion is seen to be $t \sim 0.5\,\mu s$.

where the subscript h denotes the heavy particles (their total number, mean mass and dilute-gas thermal conductivity). This expression neglects electron-ion collisions [44] and electron-electron collisions [45], both of which shorten the electron mean free path and so reduce κ_{e^-}; Equation 23 is therefore an overestimate.

Even assuming full Saha ionization and with $p_A = 1.3\,\text{atm}$, such that N_{e^-} (and thus κ_{e^-}) is overestimated, the implementation of Equation 23 only decreases the peak temperature by 7.2 % and increases the peak pressure by 2.4 %. Because the peak temperature of a default-parameter run (Section 2) with $p_A = 1.3\,\text{atm}$ is 25.8 kK, rough upper bounds for the uncertainty caused by the neglect of electron thermal conductivity in the peak temperature range from 10 to 25 kK are about 7 % (temperature) and 2 % (pressure).

4.5 Simulation in water compared to experimental data

We now proceed to test the simulation by comparison to experimental data insofar as possible. No measurements of this kind exist for NaCl, so we compare with water instead. Figure 5 compares $R(t)$ computed for a xenon bubble in water with the fit of Hopkins et al. [46] to their experimental measurements; the computed R_{max} is 2.6 % higher than the experimental R_{max}.

The close agreement in R_{max} and in the collapse timing indicates that the Keller–Miksis dynamics, the liquid properties and the expansion-phase heat and vapor exchange are implemented consistently. It does not test the collapse-phase physics (the covolume equation of state at high compression, chemistry, ionization and dense-gas conductivity) that determines the peak temperature.

The transient nature of the collapse means that temperature can be quite difficult to measure. However, estimates of the temperatures reached in acoustic cavitation in water range from about 6 to 20 kK [46], and for parameter sets typical of sonochemical experiments in water, this simulation generally remains in that range (e.g., the parameters of Figure 5 produce a peak temperature of about 8.6 kK). The water version of the simulation used here does not consider the chemical dissociation of the water, so temperatures are expected to be higher than in reality. This does not meaningfully affect the R_{max} comparison above because the water only dissociates during the $\sim 1\,\mu s$ collapse, not the expansion.

4.6 Error summary

From the error sources considered, at intermediate T_{max} ($\sim$ 10 to 25 kK), the peak temperatures calculated by this simulation have uncertainties of no more than about 15 % (as a generous bound, $\sim$ 10 % from Section 4.2 and $\sim$ 10 % from Section 4.4, added in quadrature). The peak pressure is uncertain by a similar amount.

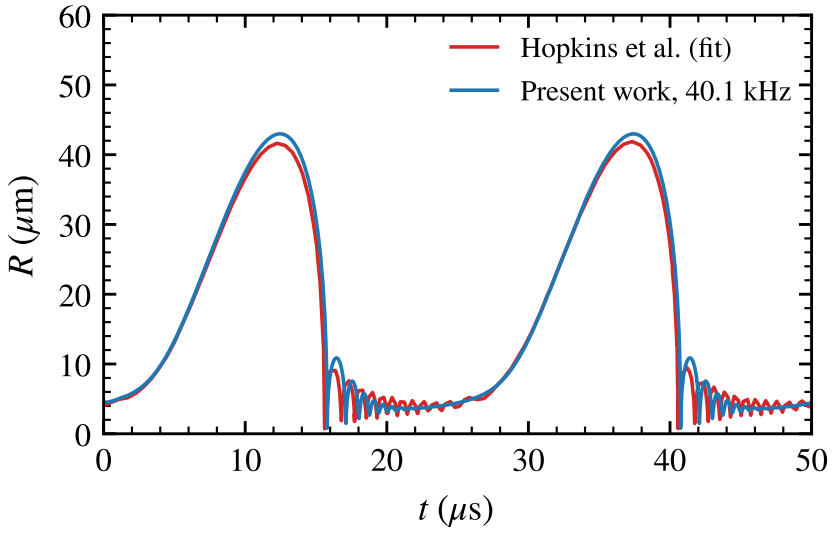


Figure 5: Comparison between the fit by Hopkins et al. to their data (red, reproduced from Ref. [46]) and the results of this simulation (blue). The spacing between successive collapses in Figure 2 of Ref. [46] corresponds to a drive frequency of 40.1 kHz, about 1.5 % above the stated 39.5 kHz. The origin of this discrepancy cannot be determined from the figure, so we compare at 40.1 kHz, which reproduces the spacing between their two collapses. Parameters: xenon and water vapor bubble, ambient temperature $T_0 = 293\,\mathrm{K}$, ambient radius $R_0 = 4.5\,\mu\mathrm{m}$, frequency $f = 40.1\,\mathrm{kHz}$, and pressure amplitude $p_\mathrm{A} = 1.5\,\mathrm{atm}$.

This error bound increases as the conditions get more intense, but in very intense collapses, bubble shape instability is expected to dominate over the errors discussed here, as explained in Section 4.2. Instability is not calculated in this work, and the pressure amplitude at which it begins to set in cannot easily be estimated from values for water. Regardless, from the discussions in Sections 4.2 and 4.4, results above $T_\mathrm{max} \approx 25\,\mathrm{kK}$ should be treated with increasing caution.

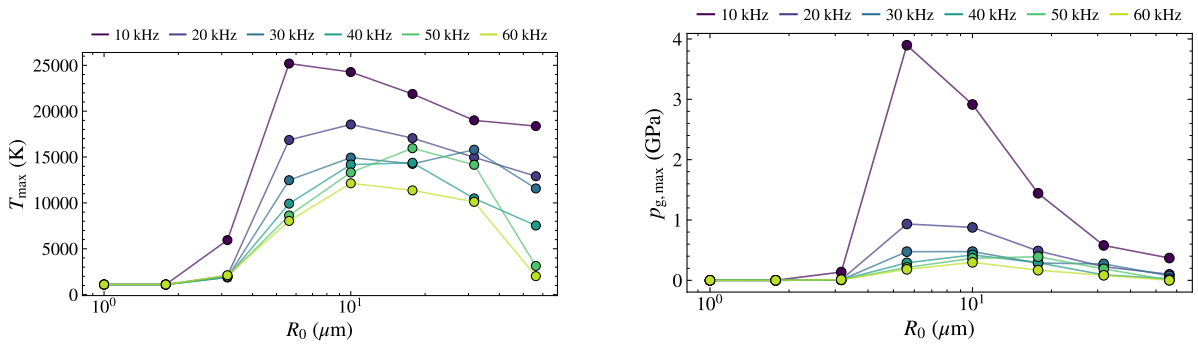
5 Results

5.1 Temperature response in various regions of parameter space

Figure 6 shows three main trends. The first is the variation of peak temperature and pressure with R_0. Notably, at low R_0, the bubbles do not undergo violent collapse; $T_\mathrm{max} \approx T_0$. This is because the Laplace pressure $2\sigma/R_0$ is large enough to resist expansion under the applied pressure amplitude, so the bubble only oscillates weakly about R_0 with negligible gas compression. At high R_0, the bubbles have a lower expansion ratio R_max/R_0 than at intermediate R_0, so the inertial collapse is not as violent. It should be noted that multiple melts of a single compound can have drastically different bubble radius distributions depending on how they are prepared (degassing, gas injection, etc.); Ref. [6] used synchrotron X-ray imaging to find the bubble radius distribution in an Al-10Cu melt sonicated with 30 kHz ultrasound. Their distribution spanned $\sim$ 10 to 100 μm. These are instantaneous sizes in a different liquid, not ambient radii, but they suggest a range of R_0 worth exploring.

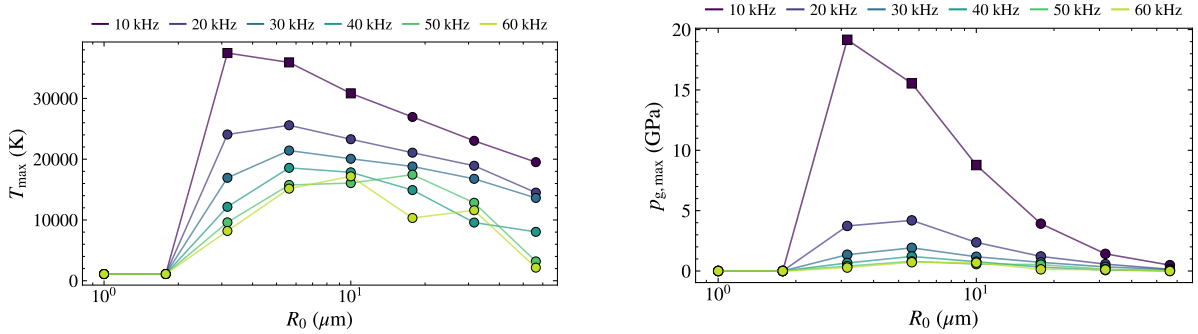
The second is the variation of T_max and the peak gas pressure $p_{g,\mathrm{max}}$ with frequency. Lower frequency gives more time for the bubble to expand, meaning that it reaches a higher expansion ratio and the collapse is more violent. In water, this effect is dampened by the high vapor pressure: though the expansion ratio increases, much more water vapor diffuses into the bubble. When the bubble collapses, much of this water vapor is trapped inside and acts as an energy sink (dissociation and high heat capacity). In low-vapor-pressure melts such as NaCl, this effect is much less significant than it is in water.

The third is the variation of T_max and $p_{g,\mathrm{max}}$ with pressure amplitude. Predictably, a higher pressure



(a) Temperature, $p_A = 1.2\,\text{atm}$

(b) Pressure, $p_A = 1.2\,\text{atm}$



(c) Temperature, $p_A = 1.3\,\text{atm}$

(d) Pressure, $p_A = 1.3\,\text{atm}$

Figure 6: Variation of peak temperature and pressure with frequency, radius, and pressure amplitude. Square data points indicate a peak temperature greater than 30 kK, so the results should be treated with caution.

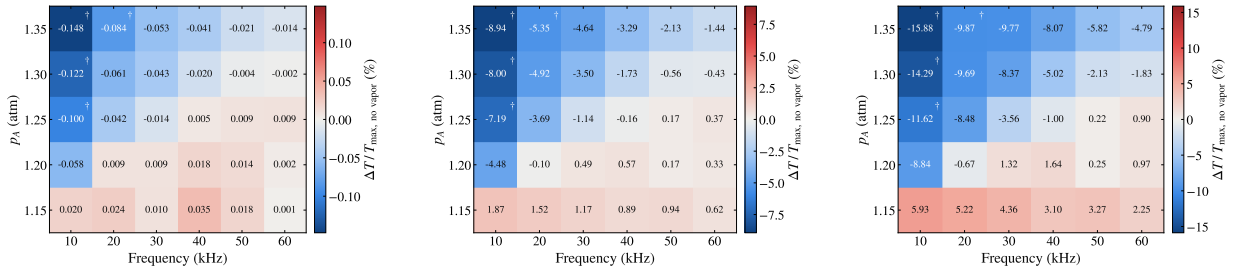
amplitude creates a more violent collapse. To summarize, intermediate bubble radius, low frequency, and high pressure amplitude contribute to more violent collapses.

5.2 Comparison to vapor-free version

The full model can only simulate NaCl; adapting its chemistry to another compound would require substantial programming and molecular data. For melts with low vapor pressures, however, the vapor can be neglected.

Thus, a vapor-free version of the simulation may be used for any compound with sufficiently low vapor pressure. Figure 7 demonstrates that for intermediate collapse conditions, the vapor is negligible (relative change in $T_{max} \lesssim 10\,\%$) for saturation vapor pressures $p_{sat} \lesssim 100\,\text{Pa}$. For lower frequencies and higher pressure amplitudes (which draw more vapor into the bubble), this deviation increases, but the more violent collapses created by these conditions mean that other error sources, such as the lack of electron impact ionization and the uncertainty in whether the bubble is stable, dominate.

Interestingly, in NaCl in particular, this approximation may remain valid for higher vapor pressures than it does in general. Vapor diffusing into the bubble allows it to expand to a higher R_{max}, which in turn allows more work to be done on the bubble and contributes to an increased temperature. However, the vapor in the bubble during collapse acts as an additional energy sink; the balance between these two phenomena is seen in Figure 8. For NaCl, these two effects happen to largely cancel out until $p_{sat} \gtrsim 3\,\text{kPa}$. In general, this cannot be assumed, so a more conservative 100 Pa bound is given for the validity of the no-vapor model. For vapor pressures beyond this value, the vapor-free model does not fail abruptly, but its error becomes increasingly uncertain.



(a) $p_{\text{sat}} = 1\,\text{Pa}$ (b) $p_{\text{sat}} = 83\,\text{Pa}$ (NaCl at 1100 K) (c) $p_{\text{sat}} = 300\,\text{Pa}$

Figure 7: Relative change in peak collapse temperature, $\Delta T_{\max}/T_{\max,\,\text{no vapor}}$, caused by melt vapor with full chemistry, as a function of driving frequency and amplitude p_A. $T_{\max,\,\text{no vapor}}$ is the peak temperature of a vapor-free run at the same (f, p_A). Each panel has its own color scale. †: $T_{\max} > 30\,\text{kK}$, where the chemistry is held at the upper limit of the kinetics model. Above 30 kK, phenomena like continuum lowering and second ionization begin to matter more, so we hold the chemistry there rather than extrapolating.

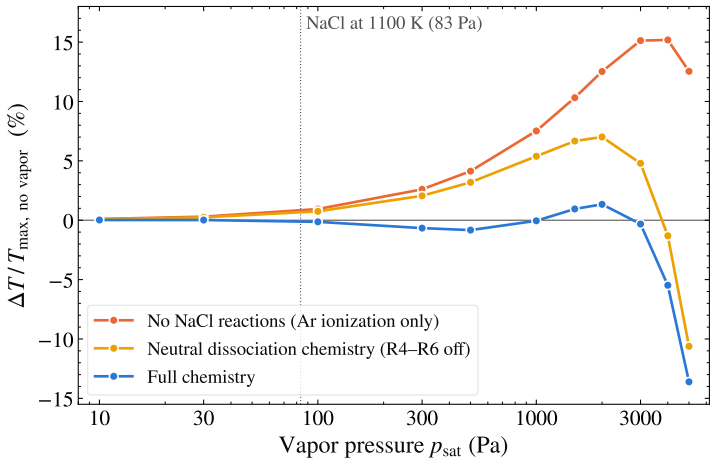


Figure 8: Deviation of peak temperature from the $p_{\text{sat}} = 0$ case as vapor pressure is increased under the default parameters (Section 2).

6 Conclusion

To summarize, we presented a model of acoustic cavitation in NaCl and tested it as far as available data allow. From the error sources we could quantify, the peak temperature and pressure are uncertain by up to about 15 % for $T_{\max} \lesssim 25\,\text{kK}$. The potential for shape instability is higher during more violent collapses [1], so the more violent the collapse, the lower the confidence in the simulation's results. It should also be noted that peak temperatures in multibubble acoustic cavitation may be lower than the peak temperatures reported here (single-bubble) due to interactions between the bubbles [47].

For compounds with low vapor pressures, we showed that a general vapor-free version of the simulation may be used. This gives an order-of-magnitude estimate of the peak temperatures and pressures in bubbles of particular radii; for sufficiently low vapor pressures, the error bounds for the vapor-free version will be comparable to those for the vapor version. The simulation can also be used to investigate how the collapse conditions vary between bubbles of different radii, which may help identify mechanisms for sonocrystallization. For example, if 10 µm bubbles give a very strong response, while 100 µm bubbles give a comparatively weak response, a reasonable hypothesis might be that the 10 µm bubbles have a larger role in crystallization.

Finally, the p_A referred to in this document is the acoustic pressure amplitude at the bubble, not the nominal output of the sonotrode, and is rarely known precisely [48]. The peak temperature is very sensitive

to it: raising p_A from 1.2 to 1.3 atm (8 %) raises T_{max} from 15.4 to 25.8 kK (67 %). The model is therefore most useful for comparing trends, such as between bubble sizes or frequencies, rather than for absolute predictions.

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